

BAHI

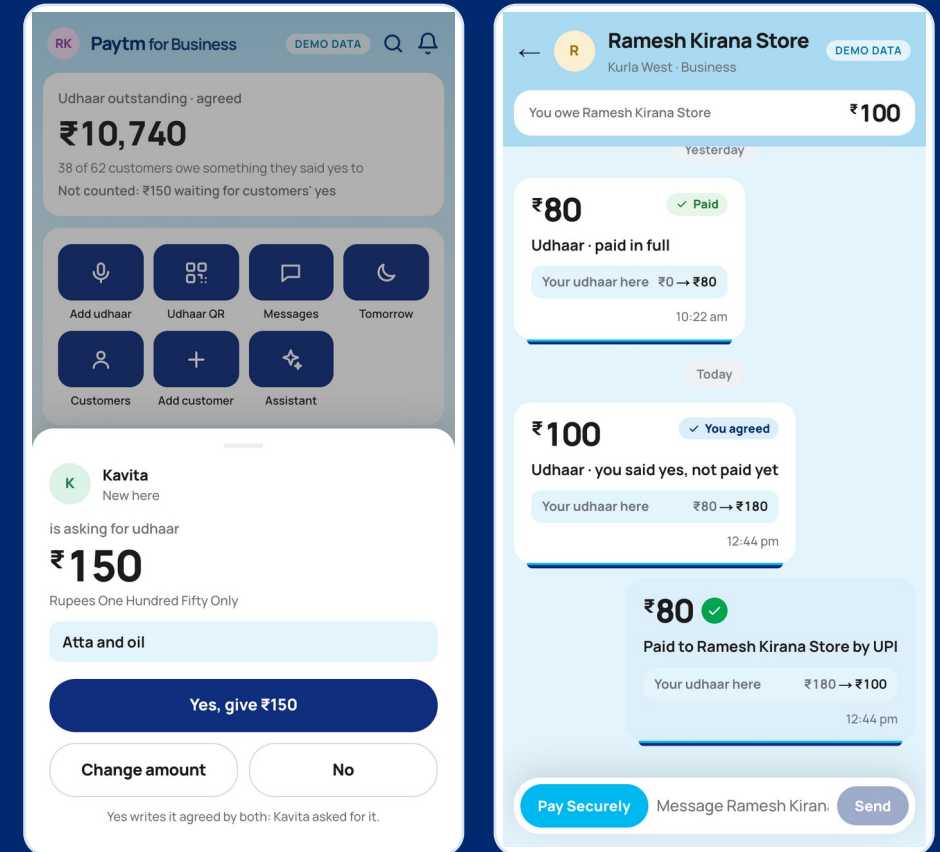
The khata both sides can see.

The customer scans the shop's udhaar QR, or asks for udhaar on her own phone. The shopkeeper tells his AI munshi the amount, and nothing is written until both sides say yes. Every night, plain code decides who actually needs a reminder.

Paytm Build for India AI Hackathon · Mumbai, 3 October 2026

Track 2: AI-Powered Financial Journeys

Team Hustlers · picked by the Paytm panel as a top 10 project



Shopkeeper · Customer · our working app, demo data

More revenue from merchants and customers Paytm already has.

₹9

**a month, added to the
Soundbox plan the merchant
already pays.**

The customer pays nothing. There is no new device to ship and no new app to install.

Our proposed price. The numbers behind it are on slide 11.

1

Customer acquisition is already done

1.57 Cr storefronts run a Paytm Soundbox, and their customers pay with Paytm. BAHl is sold to people Paytm has already won.

2

Paytm Business Khata, made two-sided

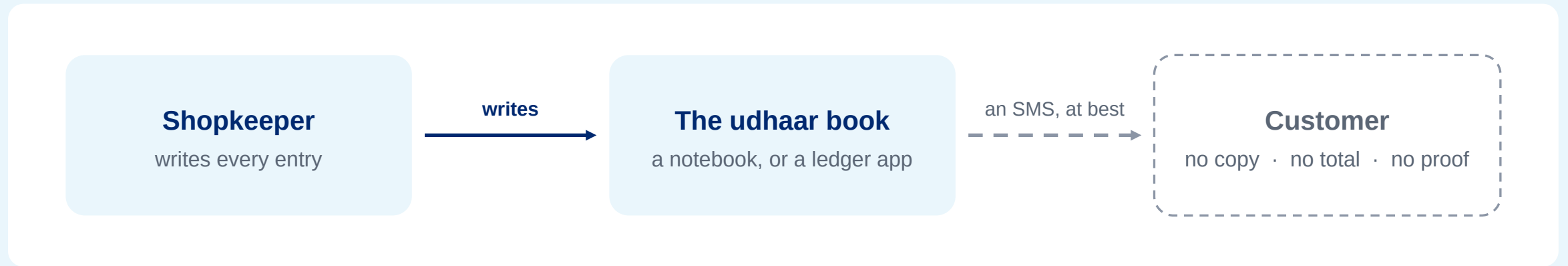
Business Khata already keeps the shop's udhaar. BAHl adds the customer's own copy, the AI munshi and the Soundbox voice.

3

Udhaar moves from cash to Paytm UPI

Every repayment is a tap in the customer's chat with the shop, paid by Paytm UPI.

The udhaar book has one author. The customer never sees it.



He can't see patterns

Someone who always pays on day 9 and someone quietly drifting away look the same in a notebook. So everyone is chased the same way.

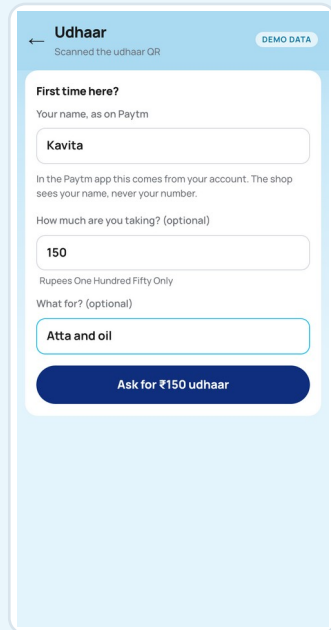
The customer has no record

He can't see what he owes across shops. If the book says ₹500 and he remembers ₹300, he has nothing to show.

Apps went digital on one side only

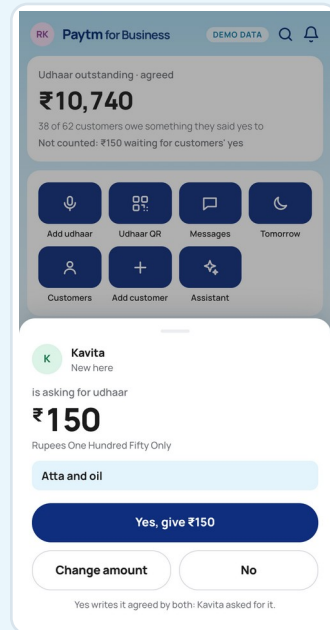
Khatabook alone has 5 Cr+ downloads. Its customers still get only an SMS — no app of their own, and no say in the entry.

One entry, agreed by both sides, inside the Paytm they already use.



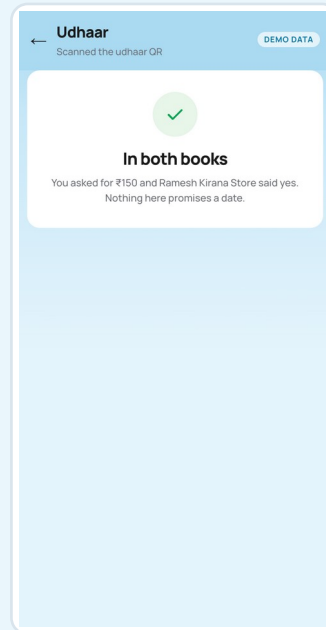
1 She scans, and can ask

A separate udhaar QR. She may say how much, and what for.



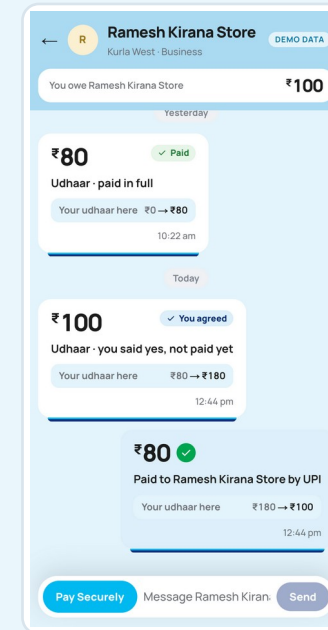
2 The shop says yes

Or says the amount to the munshi. Nothing is written before his yes.



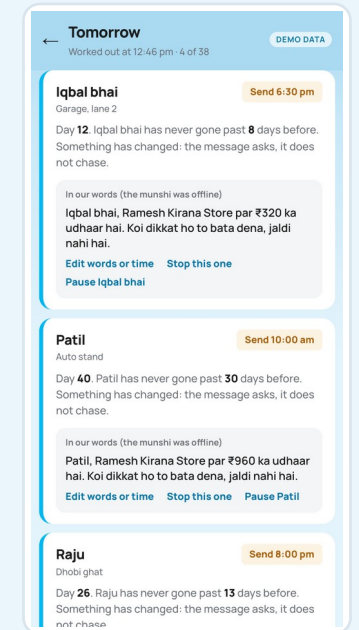
3 In both books

Her ask was her yes. Otherwise: "Yes, I owe ₹200" on her phone.



4 A chat like a passbook

Every udhaar and payment shows the total before and after it.



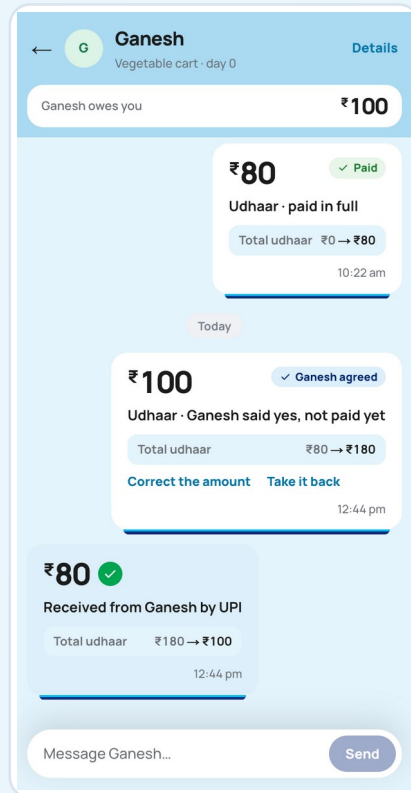
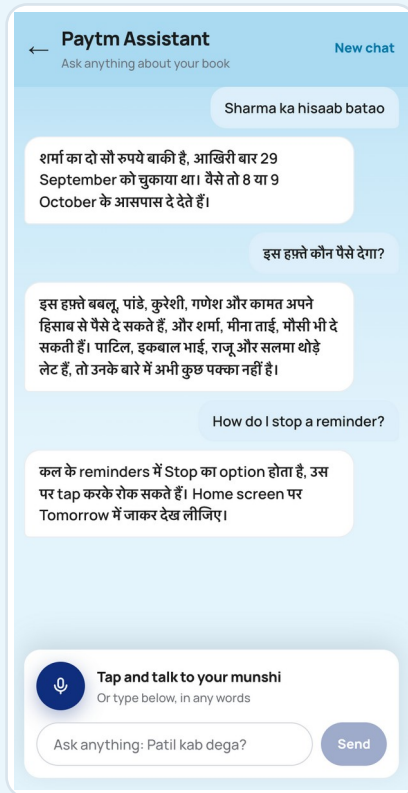
5 Code picks who hears

Only people past their own longest gap. Everyone else is held, with why.

Five jobs for AI: hear, understand, speak, remember, predict.

THE JOB	MODEL	WHAT IT DOES	WHERE YOU SEE IT
Hear	Sarvam Saaras v4	Speech to text in whatever language is spoken, with the shop's customer names as hints.	Add udhaar by voice, the munshi
Understand	Sarvam-105B	The munshi: works out what he meant and calls our tools to find a customer, put an entry on a card or look something up.	Paytm Assistant, voice entries
Speak	Sarvam Bulbul v3	Says amounts on the Soundbox and reads the munshi's replies aloud.	Soundbox, Assistant
Remember	Cognee + pgvector	Turns notes, promises and nicknames into a knowledge graph per shop, and recalls them by meaning.	"Salary on the 7th", "Pappu is Prakash"
Predict	Pattern engine + Sarvam-105B	When each customer will likely pay, and who is drifting. The munshi says it as a guess.	Customer card, Tomorrow, "इस हफ्ते कौन देगा?"

An AI munshi that does the work, and never writes without a yes.



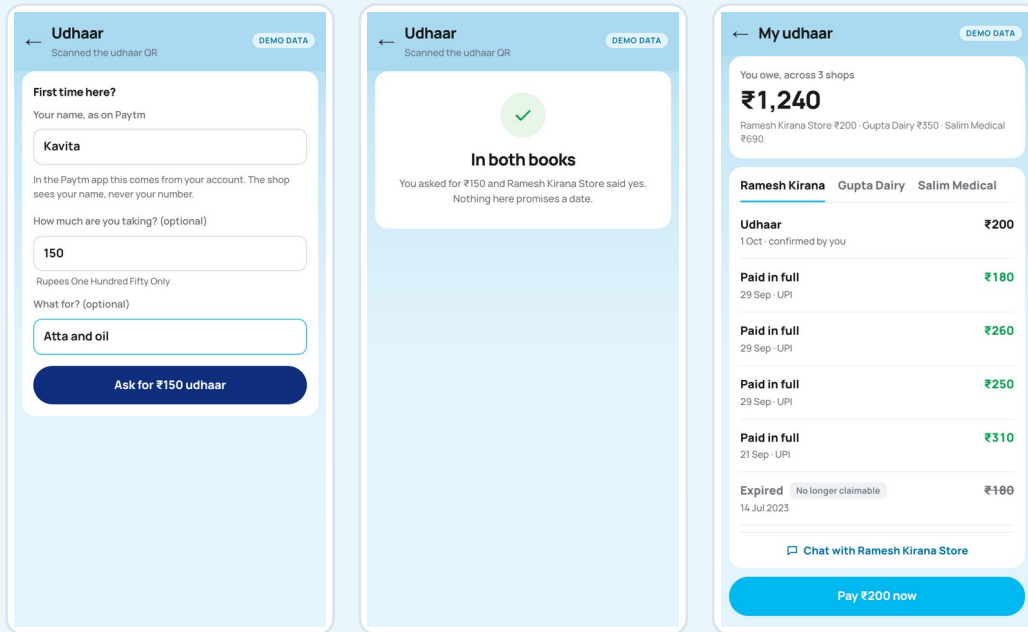
What it does

- Talk or type, in Hindi, English, Marathi or a mix. Sarvam hears any of them.
- Finds the customer by the sound of a name, or how the shop describes them.
- Asks before reading names; then says only the names.
- Puts every change on a card: an entry, a payment, a correction, taking one back, new details, a message. Only his yes writes it.
- Paytm Assistant: ask anything about the book. The chat is kept.

Its tools are our code

find_customer · counter · customer_card · propose_entry · propose_correction ·
propose_removal · propose_details · propose_message · remember · recall ·
expected_payments · tonight

Her own copy of every udhaar, inside the Paytm app she already has.

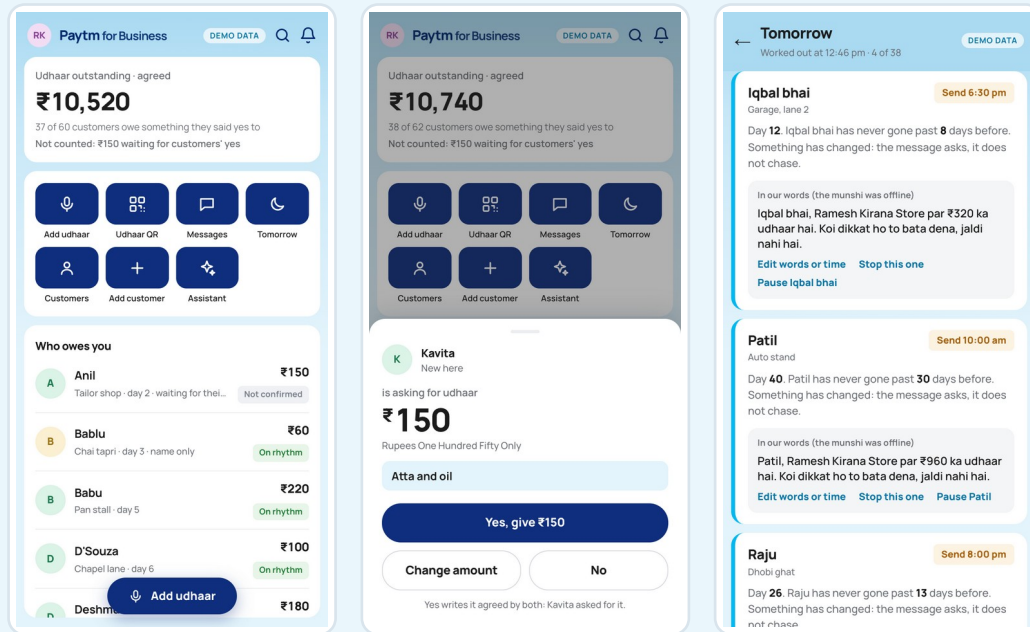


Ask · Agreed · My udhaar

What the customer can do

- Scan the shop's udhaar QR. Nothing to download: it is Paytm.
- Ask for udhaar herself: how much, and what for.
- Say yes to each entry, or tap “Not mine” or “Wrong amount”.
- A chat with each shop that reads like a passbook: the total before and after every entry.
- See what she owes across every shop, and pay by UPI, all or part.
- Reminders come privately, in her language, with no due date and no interest.

He says it once. The munshi, the Soundbox and the code do the rest.



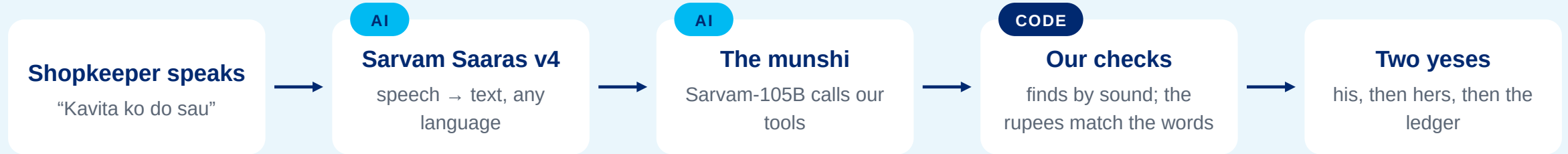
Home · A customer asks · Tomorrow

What the shopkeeper can do

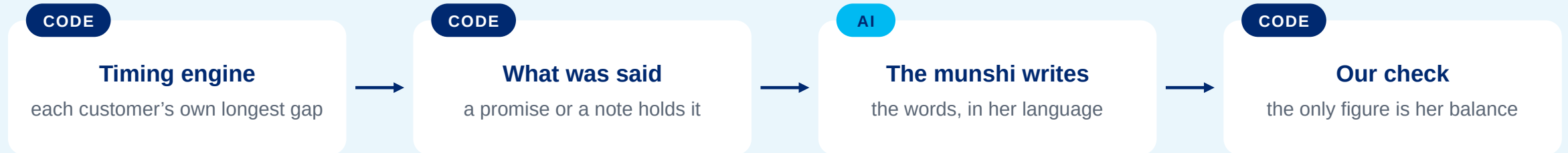
- Say the amount to the munshi in any language, or tap the keypad.
- A customer's ask pops up on his screen: Yes, No or Change amount.
- The Soundbox says the amount aloud, and never a name.
- Correct an entry or take it back, in the open.
- Ask anything: “कितना बाकी है?”, “इस हफ्ते कौन देगा?”
- BAH! remembers promises and notes, and holds a reminder for them.
- Only people late by their own pattern get a reminder. He can rewrite, move, stop or pause it.

AI understands the words. Plain code decides every number.

DURING THE DAY — RECORDING AN ENTRY



EVERY NIGHT — DECIDING WHO HEARS FROM US



How we prove it: a test fails the build if the core logic imports anything outside Python's standard library. The model only ever returns words.

STACK

Next.js 16

React 19

FastAPI

Python 3.13

PostgreSQL

Sarvam Saaras

Sarvam-105B

Bulbul v3

Cognee

Vercel · Render · Supabase

Every udhaar app serves the shopkeeper. We add the customer.

	DOWNLOADS	CUSTOMER HAS AN APP	CUSTOMER CONFIRMS ENTRIES
Khatabook	5 Cr+	No — SMS / WhatsApp	No
OkCredit	1 Cr+	No — “They don’t need the app”	No
Paytm Business Khata	Inside Paytm for Business	No — SMS with a pay link	No
Done	100+	Yes	Yes — by OTP
BAHI	—	Yes — inside the Paytm app	Yes — one tap

Two-sided

An entry becomes a fact only when the customer confirms it.

Scan, then speak

The QR says who, his voice says how much. Nothing to type.

Timed per person

Reminders follow each customer’s own repayment pattern.

No new hardware, and ₹9 more a month from each merchant.

Easy to build and sell

- Software only. No new hardware: the Soundbox already says amounts aloud.
- Built on what Paytm has: the Paytm app, its chats, UPI, the Soundbox and Business Khata.
- Sold as an add-on line on the merchant’s existing Soundbox bill, through the Paytm for Business app and the field team.
- A standalone ledger app lost ₹116 Cr in FY24 (Khatabook). BAHl needs no app of its own and no new customers.
- We built it working, end to end, during the hackathon.

₹90 → ₹99

a month from the same merchant, with no hardware delivered. Paytm’s Soundbox plan earned about ₹90 a device a month in Q4 FY24.

Our guesstimate: merchants who add BAHl

Out of 1.24 Cr merchants paying for a Paytm device, at ₹9 a month.

TAKE-UP	MERCHANTS	NEW REVENUE A YEAR
5%	6.2 lakh	₹6.7 Cr
10%	12.4 lakh	₹13.4 Cr
25%	31 lakh	₹33.5 Cr

Not counted: udhaar repayments moving from cash to Paytm UPI.

Fewer reminders, sent only to people who are actually late.

4 of 38

customers who owe get a reminder tonight.

The other 34 are not late by their own pattern, so they hear nothing.

Our demo shop: 60 customers, six months of synthetic data.

For the shopkeeper

Stops chasing people who always pay.

Spots the ones who are drifting.

Records by voice, so the queue keeps moving.

For the customer

Sees his total across every shop.

Has proof of every entry.

Is reminded privately, not at the counter.

For Paytm

Udhaar settled in cash moves to UPI.

A daily reason to use the Soundbox.

A new reason to open the Paytm app.

Four things BAHF refuses to do, each from Indian law.

1

No due dates

Customers tap “Yes, I owe ₹200” — never “I’ll pay by the 10th”.

Why: A dated promise can be a promissory note, which the IT Act does not cover.

IT Act 2000, s.1(4) & First Schedule

2

No fees, no interest

Never a rupee above the price. There is no fee field anywhere.

Why: Money-lending laws define a loan by the interest it carries.

e.g. Karnataka Money Lenders Act 1961, s.2(9)

3

No debt kept alive forever

We never ask a customer to re-confirm an old debt just to keep it alive.

Why: Each written acknowledgment restarts the 3-year limit.

Limitation Act 1963, s.18

4

No shaming, no pressure

No defaulter lists, no public alerts, no “demand” button.

Why: Public labels risk defamation; repeated nudges for gain are “nagging”.

BNS 2023, s.356 · CCPA Dark Patterns Guidelines 2023

Tested on 1,338 recordings, and running end to end.

2.1%

of spoken entries would have reached a phone wrong, and that one still needs the customer's own yes.

73.4% right from the words alone

16.9% one tap to pick who

7.6% said again

1,338 recordings: 42 lines, 12 voices, quiet, street noise and a second customer talking; 20 customers named Anubhav.

Tested

367 API tests: the schema's rules on real Postgres, the arithmetic with no database.

32 web tests: no amount typed into a screen, no shaming words, one palette.

An agent eval: another model plays the shopkeeper with a hidden goal.

Running

Next.js on Vercel, the API on Render, the ledger on Supabase, all in Singapore.

Row level security on every table: only our API reads the book.

Demo data is synthetic and labelled. The Paytm account lookup and the UPI PIN are simulated.

Code: github.com/Uchiha-Itachi0/PAYTM-BAHI

The live demo link is in our submission.

Where every fact in this deck comes from.

MARKET AND PRODUCTS

Khatabook — 5 Cr+ downloads; customers get SMS / WhatsApp

[Google Play, checked 23 Sep 2026](#)

OkCredit — 1 Cr+ downloads; “They don’t need the app”

[Google Play and okcredit.in, checked 23 Sep 2026](#)

Done — 100+ downloads; entries confirmed by OTP

[Google Play, checked 23 Sep 2026](#)

Paytm Business Khata — customer reminders by SMS with a pay link

[business.paytm.com, checked 23 Sep 2026](#)

Paytm Soundbox “deployed at 1.57 Cr storefronts”

[Paytm Q1 FY2027 earnings release, 20 Jul 2026](#)

Khatabook FY24 — ₹102.7 Cr revenue, ₹116 Cr loss

[YourStory, Nov 2024](#)

Paytm device subscription ≈ ₹90 a device a month (Q4 FY24)

[Business Standard, 22 May 2024](#)

1.24 Cr merchants paying for a Paytm device (Mar 2025)

[Paytm Annual Report FY25](#)

LAW

Information Technology Act 2000, s.1(4) and First Schedule, entry 1 (as amended 2022)

Negotiable instruments other than cheques are outside the Act.

Karnataka Money Lenders Act 1961, s.2(9)

A “loan” is an advance at interest, of money or in kind.

Limitation Act 1963, s.18

A written acknowledgment starts a fresh limitation period.

Bharatiya Nyaya Sanhita 2023, s.356

Defamation.

CCPA Guidelines for Prevention and Regulation of Dark Patterns 2023, Annexure 1, item 10

“Nagging”.